

SOFTWARE REQUIREMENTS SPECIFICATION

Inventory Management System

1. Introduction & Overview

1.1 Project Purpose

The purpose of this project is to design and develop a digital management system for small-to-medium retail businesses in the building materials sector. The system aims to replace manual, paper-based operations by automating billing, GST calculation, inventory tracking, and customer record management.

The project focuses on improving operational efficiency, reducing calculation errors, minimizing data redundancy, and providing structured access to product and sales information.

1.2 System Overview

The system is a web-based retail management platform that automates:

- Billing & Invoice Generation
- GST Calculation
- Inventory Management
- Customer Record Management
- Product Catalog Management

The system improves accuracy, efficiency, and accessibility compared to traditional manual workflows.

2. User Roles & Characteristics

2.1 System Actors

1. Admin (Owner)

- Full system access
- Manage products and inventory
- View reports
- Manage users
- Configure GST rates

2. Staff (Billing Operator)

- Generate new sales invoices
- Search products
- Add customer details
- Update stock (limited access)
- View invoice history

3. Viewer (Accountant)

- View invoices and reports
- Export GST reports
- Cannot modify data

3. Core Features Overview

3.1 Billing Management (BM)

- Generate new sales invoice
- Automatic GST calculation
- Print/download invoice
- Cancel invoice

3.2 Inventory Management (IM)

- Add/edit/delete products
- Track stock levels
- Auto stock update after sale
- Category management

3.3 Customer Management (CM)

- Add new customer
- Maintain GSTIN details
- View purchase history
- Search customer records

3.4 Reporting & Analytics (RA)

- Daily sales report
- Monthly sales report
- GST summary report

- Stock report

4. Functional Requirements

ID	Requirement Title	Description	Input	Process	Output	Priority
1	Generate New Sale Invoice (Billing Management)	Staff generates a new sales invoice for a customer.	Customer details, Product list, Quantity	Fetch product price; Calculate subtotal; Apply GST automatically; Generate total amount	Invoice generated with unique invoice number; Stock automatically updated	P0
2	GST Calculation (Billing Management)	System automatically calculates applicable GST (CGST, SGST, IGST).	Product price, GST %	$GST = (Product\ Price \times GST\ \%)$	Accurate tax breakdown displayed on invoice	P0
3	Add Product (Inventory Management)	Admin adds new product into the system.	Product name, category, price, GST %, stock quantity	Validate no duplicate product names	Product stored in system	P0
4	Auto Stock Update (Inventory Management)	Stock quantity automatically reduces after each sale.	Sale transaction data	Deduct sold quantity from available stock	Updated stock value	P0

5	Add Customer (Customer Management)	Add new customer details into the system.	Name, phone number, GSTIN (optional), address	Validate phone number format	Customer details saved	P0
7	Generate Sales Report (Reporting & Analytics)	System generates monthly sales reports.	Date range selection	Aggregate sales data	Report viewable and exportable (PDF/Excel)	P1

5. Non-Functional Requirements

5.1 Performance

- Page load time < 5 seconds
- API response time < 700 ms
- Support up to 500 concurrent users

5.2 Scalability

- Support up to 8,000 products
- Allow multi-user access

5.3 Availability

- 95.5% uptime SLA
- Maintenance during non-business hours

5.4 Security

- Role-based access control
- Encrypted passwords
- Secure login authentication
- Data protection for GST records